

COE817 Lab 2

Objectives

- In this lab, you will familiar yourself with Java cryptography class libraries and work on authentication protocols.
- **Two students** will work as a group.
- Duration: **two weeks**.

Preparing for Lab 2:

Familiar yourself with Java Cryptography Architecture(JCA). Especially the ciphers used in this lab including DES, RSA, and digital signature.

JCA includes built-in providers for many of the most commonly used crypto-graphic algorithms, including RSA, DSA, and ECDSA signature algorithms, the DES, AES, and ARCFOUR encryption algorithms, the MD5, and SHA-256 message digest algorithms (hash functions), and the Diffie-Hellman key exchange algorithms.

Java's security and cryptography classes are divided into two main packages: `java.security.*` and `javax.crypto.*`. The `javax.crypto.*` package is collectively referred to as the Java Crypto Extensions or the JCE. Classes for cryptographic hashing and digital signatures can be found in `java.security.*`, whereas ciphers such as DES, AES, RSA and MACs (message authentication code) are located in the JCE.

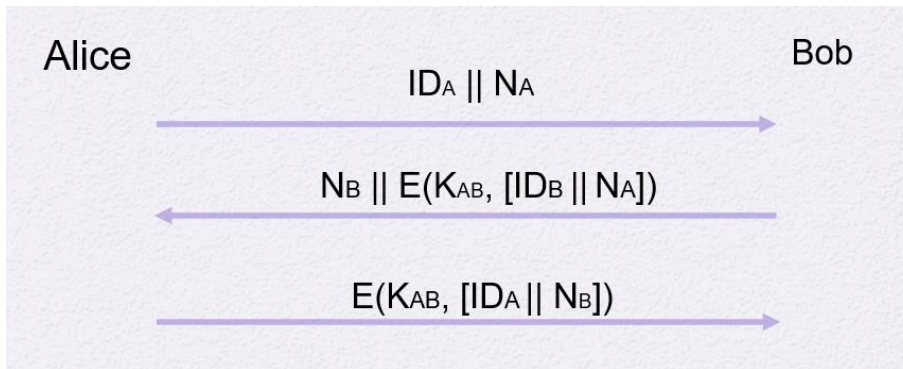
You could use JCA reference guide to familiar yourself with the `java.security.*` and `javax.crypto.*` packages

<https://docs.oracle.com/en/java/javase/21/security/java-cryptography-architecture-jca-reference-guide.html#GUID-2BCFDD85-D533-4E6C-8CE9-29990DEB0190>

Your tasks

Project 1: Symmetric key-based authentication protocol

Write a Java socket program to implement the symmetric key-based authentication protocol that is explained in class.



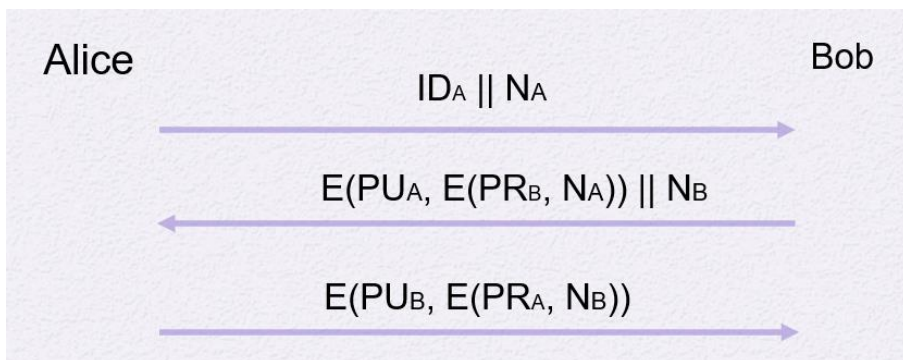
Use DES or AES to encrypt and decrypt messages.
Pick up your own ID, nonces and symmetric keys.

The detail description of the protocol is as follows:

- (1) Alice sends its identity and a nonce N_A to Bob
- (2) B responds a nonce N_B and an encrypted message that includes B's identity and nonce N_A . K_{AB} is the key.
- (3) A sends B an encrypted message that includes A's identity and nonce N_B .

Project 2: Public key-based authentication protocol

Write a Java socket program to implement the following public key-based authentication protocol.



- You need to
- (1) Generate RSA public/private key pair
 - (2) Encrypt and decrypt message using RSA
 - (3) Pick up ID and nonces by yourself.

Project 3: Digital signature

Alice(A) sends Bob(b) a message M with its signature.

$$A \rightarrow B: M \parallel \text{Sig}_A(M)$$

Where $\text{Sig}_A(M)$ means the digital signature of M generated by A

Digital signature generation and verification methods are provided by Java cryptography class libraries. The program should create an RSA key pair and then signs message M and displays the signature. Finally, verify the signature with the corresponding public key.

If the signature is verified successfully, the receiver Bob will be able to authenticates Alice (digital signature supports sender authenticity). However, this protocol is vulnerable to replay attack.

Modify the authentication protocol to eliminate this replay attack.

Write Java program to implement your revised protocol.

What you needed to demonstrate to your TA

For project 1, show the following messages::

- the received message 1
- the received message 2
- the decrypted message 2
- the received message 3
- the decrypted message 3

For project 2, show the following messages::

- the received message 1
- the received message 2
- the decrypted message 2
- the received message 3
- the decrypted message 3

For project 3, explain your revised design of the protocol. Show the received signature of message M.

Submitting your lab

You must submit your source codes of project 1, 2 and 3 to lab2 submission folder on D2L at least 24 hours prior to the start of your scheduled lab period for Lab 3.